Kivan Polimis

+1 713-292-6693
Houston, TX
kpolimis
kivanpolimis.com
kivan.polimis@gmail.com

Data scientist (PhD) with 13+ years turning ambiguous, high-stakes questions into decisions for cross-functional product, policy, and engineering teams — from defining measurement frameworks and applying statistical methods to shipping forecasting and ML systems that change what stakeholders do. Fluent in Python, R, and SQL; experienced with cloud data infrastructure (AWS, Azure, Snowflake) and modern analytics workflows.

EMPLOYMENT

2021–Present **Senior Data Scientist,** Karna, Atlanta, GA

- Engineered and launched a document-intelligence service (Azure Document Intelligence, GPT-4, AKS) — architecture, vendor selection, CI/CD, automated testing, schema enforcement, and 5-language translation; processed >20k surveys in parallel in under a day at >99% multiple-choice and >95% field accuracy, replacing 1+ month of manual processing for a 120k-member program.
- Defined extraction accuracy, latency, and cost as evaluation KPIs; ran a head-to-head benchmark of Azure Document Intelligence, AWS Textract, and GCP Document AI; selected Azure DI on measured performance — results set the accuracy thresholds (>99% MC, >95% field) used as go/no-go launch criteria.
- Designed and ran A/B tests of message variants (n≈10,000 per wave); stratified treatment effects by hesitancy, ideology, and demographic segments using ordered logistic regression; hesitancy segment moderated message preference by 30–50%, driving differentiated ad purchasing by audience segment.
- Architected Snowflake data infrastructure and multi-source ETL pipelines (Python, R, shell scripting) to ingest and transform disparate datasets for program utilization and analytics.
- Designed a production ML pipeline for risk stratification using XGBoost and MLflow with automated drift monitoring; associated with 5% lower unnecessary emergency department visits and downstream costs.
- Evaluated a deployed maternal CVD screening workflow (n=4,485): 92% screening uptake, 3.6% positivity; 72% of at-risk patients routed to cardiology vs. 12% of unscreened patients.

Awards: President's Award (2023), Technical Achievement Award (2023), Community Engagement Award (2022), Outstanding Publication runner-up (2022), Technical Achievement runner-up (2024)

2020–2021 **Senior Policy Advisor,** Office of Harris County Judge Lina Hidalgo, Houston, TX

- Automated COVID-19 surveillance reporting by replacing hand-transcribed web data and manual multi-source joins with a reproducible R pipeline ingesting DSHS, HCPH, and national datasets, reducing processing latency by 90% and enabling real-time data quality for policymakers.

2020–Present **Principal and Founder (consulting),** Atlas Analytics, Houston, TX

- Applied NLP (TF-IDF, topic modeling, k-means clustering) to Twitter follower timelines to segment users by natural hair product affinity; automated targeted Twitter outreach to the highest-affinity cluster; reached users converted at 20% higher referral rates than untargeted users.
- Designed a Python/SQL entity resolution pipeline joining fragmented SBA datasets with merchant registries via a tiered probabilistic matching algorithm (86% recall on a manually labeled hold-out set), enabling downstream customer analytics and market segmentation.

2018–2020 **Data Scientist,** MAANA, Houston, TX

- Partnered with solutions architects, product managers, and SMEs to engineer and deploy containerized ML microservices (Docker, Kubernetes) powering a real-time vessel routing optimizer across global multi-port routes, with ingestion pipelines parsing Q88 XML and integrating live weather, war-risk, and port data to cut fleet planning from hours to minutes.

2017–2018 **Postdoctoral Researcher,** Università Bocconi, Milan, Italy

Carlo F. Dondena Centre for Research on Social Dynamics and Public Policy

- As a Bocconi Institute for Data Science fellow, built HPC-scale pipelines ingesting high-volume social media APIs (Twitter, Facebook), applied NLP to unstructured text streams, and fused API-derived signals with structured survey data in a Bayesian hierarchical model to produce high-frequency forecasts.

PROJECTS

Wandarr — Travel Intelligence & Itinerary Platform

- Launched wandarr, a production travel intelligence PWA (FastAPI, PostgreSQL, React + Vite) fusing Google Places POI discovery and drone airspace data (OpenAIP) into ranked itineraries.
- Syncs bidirectionally with TREK for day-by-day execution; account-free invite links with iOS/Android PWA install (vite-plugin-pwa); self-hosted via Docker Compose behind Cloudflare.

Transaction Text Classifier

- Developed a transaction text classification pipeline for personal finance data, evaluating TF-IDF + Logistic Regression, LinearSVC, and XGBoost against Sentence-BERT transformer architectures.
- Implemented a Majority-Vote Ensemble achieving 89.7% accuracy, with classical TF-IDF models matching transformer-based performance.

TEACHING

2021–Present **Lecturer (part-time),** University of Washington, Department of Sociology, Seattle, WA

- Developed and taught applied labs on behavioral and digital trace data analysis (Google Trends, Fitbit, sensor data), data visualization, NLP, and algorithmic fairness; challenged students to interrogate models for potential bias from training data, historical inequities, and deployment context.

2017–Present **Certified Instructor,** The Carpentries (Software Carpentry; Data Carpentry),

TECHNICAL SKILLS

Cloud/DevOps	AWS (ECR, SageMaker, Lambda, S3), Azure (ACR, AKS, Azure Functions), Docker, Git, CI/CD
Data Engineering	ETL/ELT pipeline design, distributed data systems (Snowflake, dbt, Airflow, PySpark, Databricks, BigQuery), PostgreSQL, SQL Server, SQLite, schema enforcement, data lineage, medallion architecture
Programming	Python, SQL, R, shell scripting
Project Tools	Jira, GitHub Projects

EDUCATION

2017 **Ph.D.,** Computational Social Science, University of Washington
National Science Foundation Graduate Research Fellowship recipient
Center for Statistics and the Social Sciences: Certification in Social Statistics
Center for the Study of Demography and Ecology Pre-doctoral Fellow
Data Science for Social Good Fellow

2012 **M.A.,** Sociology, University of North Carolina at Chapel Hill
Carolina Population Center Pre-doctoral Fellow

2009 **B.A.,** Sociology, Princeton University

SELECT PUBLICATIONS

2020 Alexander, Monica, **Kivan Polimis**, and Emilio Zagheni. "Combining social media and survey data to nowcast migrant stocks in the United States". *Population Research and Policy Review.*

2020 Salganik, Matthew, Ian Lundberg, Alexander T. Kindel, Caitlin E. Ahearn, **et al.**. "Measuring the predictability of life outcomes with a scientific mass collaboration". *Proceedings of the National Academy of Sciences.*

2019 Alexander, Monica, **Kivan Polimis**, and Emilio Zagheni. "The impact of Hurricane Maria on out-migration from Puerto Rico: Evidence from Facebook". *Population and Development Review.*